

Vireen Chowdary Vesangi

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🌐 Portfolio: vireen555.github.io | [GitHub](#)

SUMMARY

B.Tech in Artificial Intelligence Engineering (Robotics specialization), completing an MBA in Information Systems & Analytics. Works at the intersection of autonomous systems, applied ML, and business analysis — from simulation and computer vision to the decision-support layer that lets robotics organizations scale safely and repeatably. Ships end-to-end: recent independent work includes a deployed probabilistic simulation engine (Python/FastAPI/React) and a multi-signal computer-vision system (PyTorch, OpenCLIP, YOLOv8). Comfortable across engineering, operations, and leadership contexts.

SKILLS

Robotics & AI: ROS, Gazebo, Webots, ArduPilot, Computer Vision (OpenCV, OpenCLIP, YOLOv8, EasyOCR), Deep Learning (PyTorch, CNN, VGG, fine-tuning), Path Planning (RRT), Embedded Systems (Arduino, Raspberry Pi), BERT / NLP, local LLM deployment (Ollama, Pydantic).

Engineering & Data: Python (Pandas, NumPy, scikit-learn), SQL, FastAPI, REST API design, React, pytest, Git/GitHub, cloud deployment (Vercel, Render), ETL pipeline design, Monte Carlo simulation, empirical-Bayes / hierarchical modelling, model calibration.

Business & Analytics: Requirements gathering (BRD/FRD), stakeholder communication, KPI design, dashboarding (Power BI — DAX, Power Query, RLS; Tableau — LOD, parameters), segmentation, forecasting (Prophet), Advanced Excel (Pivot Tables, VBA), SPSS.

FEATURED PROJECT

Pilot-to-Scale Decision Support for Autonomous Robotics

independent study

- Built an end-to-end decision-support framework for autonomous robotics organizations transitioning from pilot deployments to large-scale operations, structured around a 5-stage **Pilot-to-Scale Lifecycle** with explicit go/hold/stop logic per pilot.
- Designed a **Pilot Readiness Dashboard** in Power BI on synthetic operational data (5 tables, 4 pilots across global environments) with weighted readiness scoring and fleet-level rollups.
- Prototyped a **local-LLM Robotics BA Copilot** (Python, Streamlit, Ollama) that converts unstructured operational text — weekly pilot notes, incident descriptions, customization requests — into structured artifacts with auto-split per-pilot tabs and a cross-pilot fleet-synthesis mode. Privacy-first: no operational data leaves the local machine; output schemas validated with Pydantic.
- 2-min walkthrough: [video](#) | Full study (PDF): [available on request](#)

PORTFOLIO PROJECTS

Captain's Call — Probabilistic Match Simulator (Solo, Deployed)

Python, pandas, FastAPI, React

- Built and deployed a live probabilistic T20 simulation engine end-to-end: pandas ETL over 1,244 matches / 2,95,733 ball-by-ball events (18 seasons), a per-event outcome model with **empirical-Bayes shrinkage** and multi-level hierarchical back-off (26,250-cell parameter grid per discipline), and a **Monte Carlo win-probability engine** (2,000 simulations per call).
- Diagnosed and fixed a systematic 64/36 estimation bias in the Monte Carlo layer, restoring ~50/50 symmetry for identical line-ups; calibrated the intent-control layer to MAE ~1.04pp against real data; stateless state-in/state-out FastAPI backend with a GPU-free pytest suite (invariant, monotonicity, symmetry tests).
- [Click here to play with the Simulation](#)
- [Click here to view the GitHub repo](#)

GeoSpot — Multi-Signal Country Prediction from Images

PyTorch, OpenCLIP, YOLOv8, EasyOCR

- Rebuilt a 2022 university project into a modern CV system fusing three signals: **OpenCLIP ViT-B/32** zero-shot scene classification, **EasyOCR** script detection across 8 non-Latin scripts, and a **YOLOv8 flag detector fine-tuned on synthetic flag-in-scene composites** (real flags pasted onto street backgrounds to fix a whole-image-as-flag failure mode).
- Trust-weighted fusion with per-module weights set from measured precision on a labeled test set; updating weights from empirical evidence improved top-1 accuracy from 66.7% to 73.9% (top-3: 87.0%).

Autonomous Gardening Robot

ROS, SLAM, RRT, OpenCV DNN, Arduino, SOLIDWORKS

- Designed and simulated a distributed autonomous gardening robot for urban / indoor precision agriculture: a self-navigating platform that detects potted plants, checks soil moisture, and waters plants without human intervention.
- **Navigation:** implemented autonomous exploration and path planning in ROS using SLAM (Simultaneous Localization and Mapping) for real-time environment mapping and the RRT (Rapidly-exploring Random Tree) algorithm for frontier-based path planning to unexplored regions.
- **Perception:** built an object-detection pipeline in OpenCV using a MobileNet SSD (Single Shot Detector) Deep Neural Network to identify fruits, plants, and long-range obstacles that proximity sensors cannot detect; achieved ~35 fps on a Raspberry Pi after porting from a Windows dev environment for portable, on-device inference.

- **Moisture control:** designed the moisture-detection circuit (Arduino Uno + soil sensor, prototyped in Tinkercad) with dielectric-constant-based sensing and RGB-LED state indication (red = too dry, blue = too wet, green = correct); wired the sensor to drive an electric-motor watering trigger when soil conductivity fell below threshold.
- **Mechanical design:** full CAD assembly in SOLIDWORKS — caterpillar-tracked base, on-board water tank, malleable 3D-sketched pipes, hydraulic-inspired gripper, Arduino / Raspberry Pi mount — assembled via the mate tool; taught the team the physics constraints of turning a simulation into a manufacturable design.

The Road to Biped Robot Stair Climbing: A Simulation Approach

Webots, ROS, Python, IKPy

- Simulation study of bipedal humanoid robots (NAO and Darwin) in Webots and ROS, evaluating stair-climbing feasibility on uneven terrain — a classic hard problem in legged locomotion where a shift of the centre of mass beyond the support polygon causes instability.
- Built Python control scripts to test stability parameters across step heights and gait configurations; framed the problem in terms of inverse kinematics (mapping desired foot placement back to joint angles) and gait generation (sequencing joint trajectories for balanced ascent).
- Featured in an [appreciation post by Amrita Vishwa Vidyapeetham](#) for the project's clarity and depth.

UAV Swarm Simulation with Pandemic-Response IoT — Team Project

ROS, Gazebo, SDN, IoT

- Simulated a coordinated UAV swarm in ROS / Gazebo with an integrated IoT thermal-detection module for contactless health monitoring — framed as a rapid pandemic-response deployment scenario for large public gatherings.
- Implemented **Software Defined Networking (SDN)** to manage swarm-wide communication, separating the control plane from the data plane so that flight coordination and thermal-data flow could be optimized independently as the swarm scaled.
- Simulation-first design decision: chose ROS + Gazebo (over hardware prototyping) to allow multi-agent scenarios that would be infeasible to test with physical UAVs at the time.

EXPERIENCE

Business Analyst Intern

May – Jul 2025

Symbiosys Technologies, Visakhapatnam
Operations

Summer internship — operational data & receivables analysis under VP

- Analyzed 1,500+ multi-currency invoices in Tableau, surfacing ~Rs 2.7 Cr in total overdue exposure and a single overseas market holding ~Rs 1.2 Cr in receivables; built a Python real-time currency-conversion utility (forex API with fallback) to standardize INR/JPY/SGD/AUD data.
- Used SQL and Advanced Excel to cleanse and join client/invoice datasets; revenue analysis showed one vertical (Engineering Consultancy, ~27%) driving a disproportionate share of revenue.
- Statistical analysis in SPSS: Spearman rank correlation ($\rho = -0.001$, $p = 0.989$) showed invoice size did **not** predict payment delay; K-Means clustering (3 segments) revealed elevated delinquency on 45-day terms, leading to a credit-policy recommendation presented in management reviews.

AI Intern

Feb – May 2023

Ozonetel Communications Pvt. Ltd., Hyderabad

Agent Assist AI — NLP and front-end integration

- Built the front-end web interface for an Agent Assist AI product, handling integration between LLM-driven backends and the customer-facing UI.
- Applied BERT-based NLP modelling to unstructured customer-interaction data; contributed to model-validation workflows in a production business context.

EDUCATION

MBA — Information Systems, Analytics & International Business

2024 – 2026

GITAM University, Visakhapatnam

CGPA: 7.6 / 10

B.Tech — Artificial Intelligence Engineering (Robotics Specialization)

2019 – 2023

Amrita Vishwa Vidyapeetham, Amritapuri

CGPA: 7.5 / 10

CERTIFICATIONS

- Microsoft PL-300 Power BI Data Analyst (in progress)
- Microsoft Power BI Data Analyst (Coursera)
- Microsoft Engage 2021
- Shore Fest Datathon 1st Place Finish